Optimal Execution under Market Impact

A Deterministic Comparison of TWAP, VWAP, and Almgren–Chriss

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Abstract

Executing large orders in financial markets involves a structural trade-off between execution cost and risk. This note develops the deterministic layer of an optimal execution framework and compares three canonical strategies: TWAP, VWAP, and Almgren–Chriss.

The analysis focuses on how inventory evolves over time, how execution intensity differs across slices, and how execution urgency affects the optimal trajectory. This deterministic study serves as the foundation for a subsequent stochastic analysis under Brownian price dynamics and market impact.

Contents

1. Introduction

Executing large orders is not trivial. Every trade interacts with market liquidity and may affect the execution path in a non-negligible way. This creates a fundamental trade-off between:

- **Execution cost**, driven by impact and liquidity consumption
- **Execution risk**, driven by price uncertainty while inventory remains open

Simple execution benchmarks such as TWAP and VWAP are commonly used in practice, but they do not explicitly solve the trade-off between cost and risk. By contrast, the Almgren–Chriss framework formulates execution as an optimization problem.

In this first phase of the project, the objective is to answer two core deterministic questions:

1. How do TWAP, VWAP, and Almgren–Chriss differ in their execution schedules?
2. How does execution urgency λ affect the optimal inventory trajectory?

The stochastic evaluation of cost and risk is left for the next stage of the project.

2. Deterministic Execution Strategies

2.1 TWAP

The Time-Weighted Average Price (TWAP) strategy distributes the order uniformly across time. If a total order of size Q is split into N slices, then the shares executed in each slice are:

$$n_k = \frac{Q}{N}, \qquad k = 1, \ldots, N$$

This implies a linear decay in the remaining inventory. TWAP is simple and easy to implement, but it ignores both liquidity conditions and any explicit notion of risk.

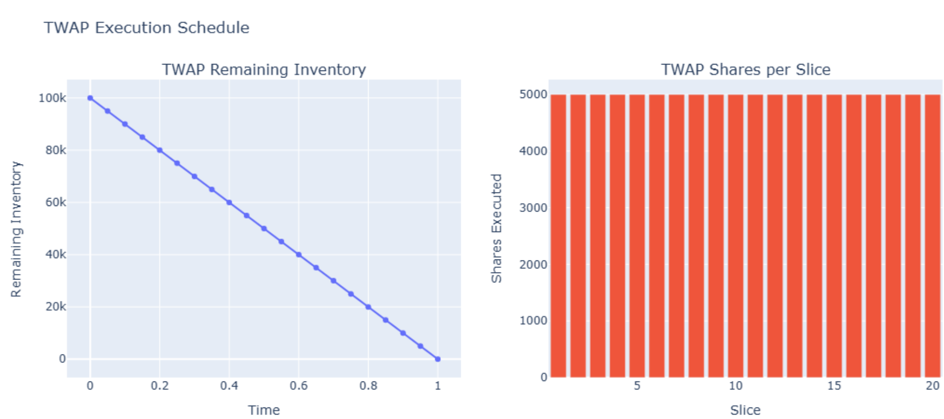

Figure 1: TWAP execution schedule: remaining inventory and constant shares per slice.

2.2 VWAP

The Volume-Weighted Average Price (VWAP) benchmark aligns execution with expected liquidity. The idea is to allocate more shares to periods where the market is expected to trade more volume.

Given an expected intraday volume profile $\{V_k\}_{k=1}^{N}$, the execution schedule is:

$$n_k = Q \cdot \frac{V_k}{\sum_{j=1}^{N} V_j}$$

Unlike TWAP, VWAP is sensitive to the expected shape of market activity. It does not optimize a cost–risk objective, but it does produce a more market-aware allocation.

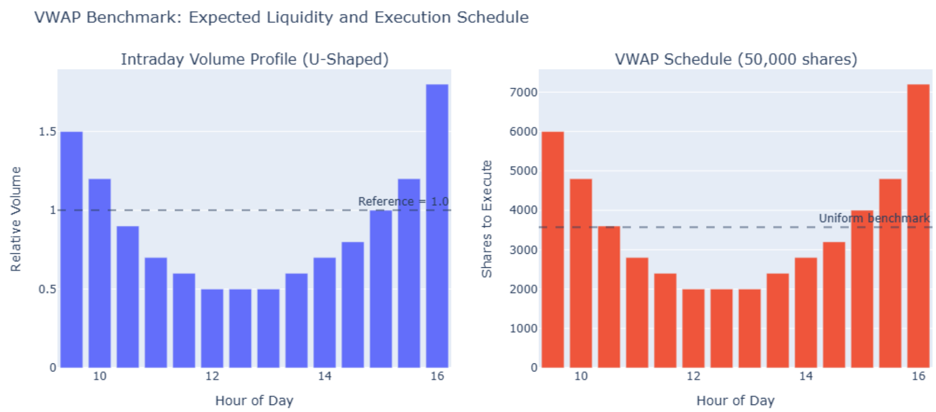

Figure 2: VWAP schedule generated from a stylized U-shaped intraday volume profile.

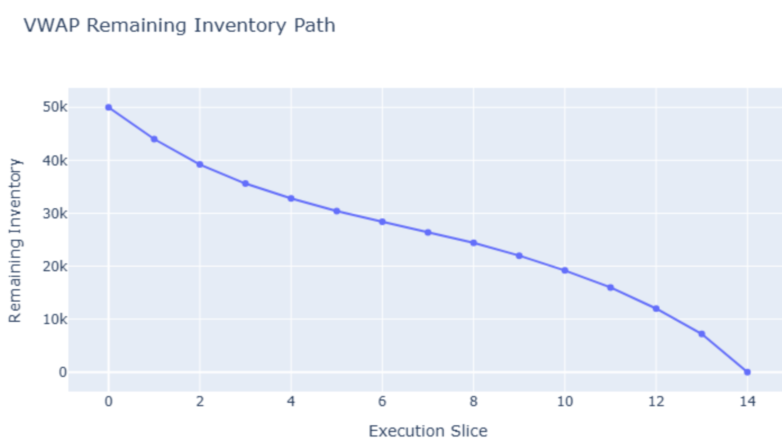

Figure 3: Remaining inventory under VWAP execution.

3. Market Impact: Execution Perspective

To motivate optimal execution, it is useful to distinguish two forms of impact.

3.1 Temporary Impact

Temporary impact arises from demanding immediate liquidity and depends on execution speed:

$$\Delta S_{\text{temp}}(t) = \eta\, v(t)$$

where η is the temporary impact coefficient and $v(t)$ is the trading rate.

3.2 Permanent Impact

Permanent impact reflects the cumulative information content of executed volume:

$$\Delta S_{\text{perm}}(t) = \gamma\,(Q - x(t))$$

where γ is the permanent impact coefficient and $x(t)$ is the inventory remaining at time t.

4. Almgren–Chriss Optimal Execution

Let $x(t)$ denote the shares remaining to execute at time $t \in [0, T]$, with:

$$x(0) = Q, \qquad x(T) = 0$$

The deterministic cost–risk objective is:

$$\min_{x(t)} \quad \eta \int_0^T \dot{x}(t)^2\, dt + \lambda\sigma^2 \int_0^T x(t)^2\, dt$$

The optimal inventory trajectory is:

$$x^*(t) = Q \cdot \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}$$

with

$$\kappa = \sqrt{\frac{\lambda\sigma^2}{\eta}}$$

As λ increases, the optimal strategy becomes increasingly front-loaded.

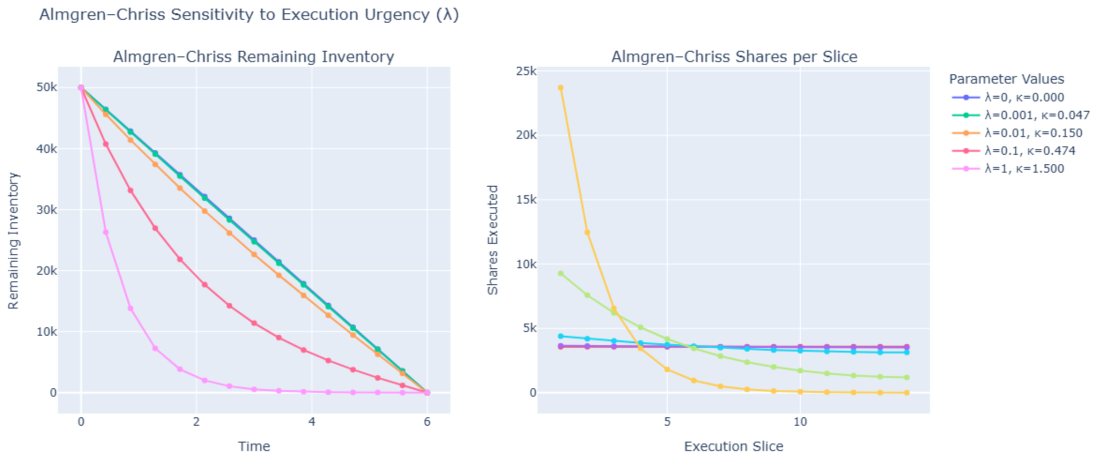

Figure 4: Sensitivity of Almgren–Chriss execution to the urgency parameter λ.

5. Comparative Analysis of Execution Strategies

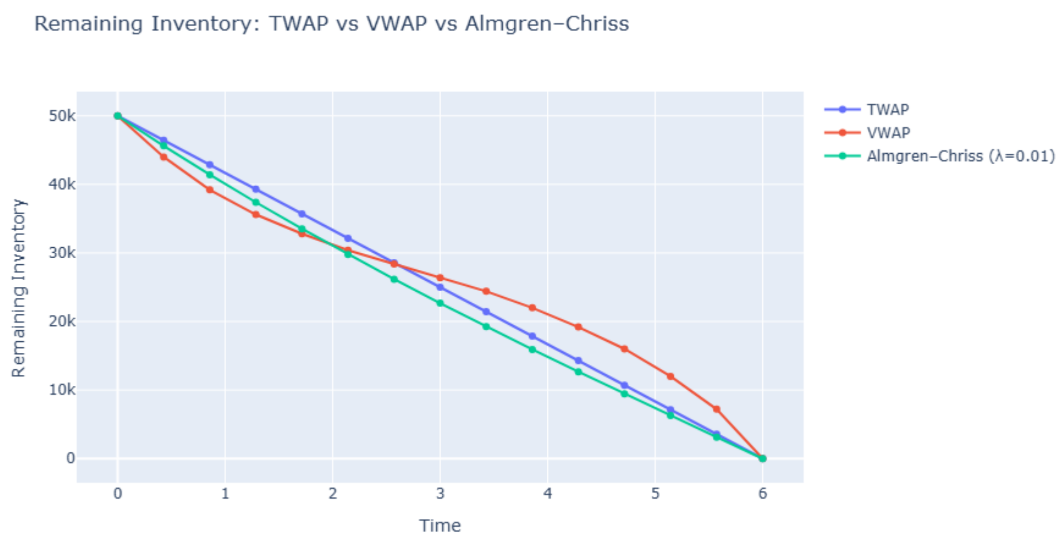

Figure 5: Comparison of remaining inventory trajectories for TWAP, VWAP, and Almgren–Chriss.

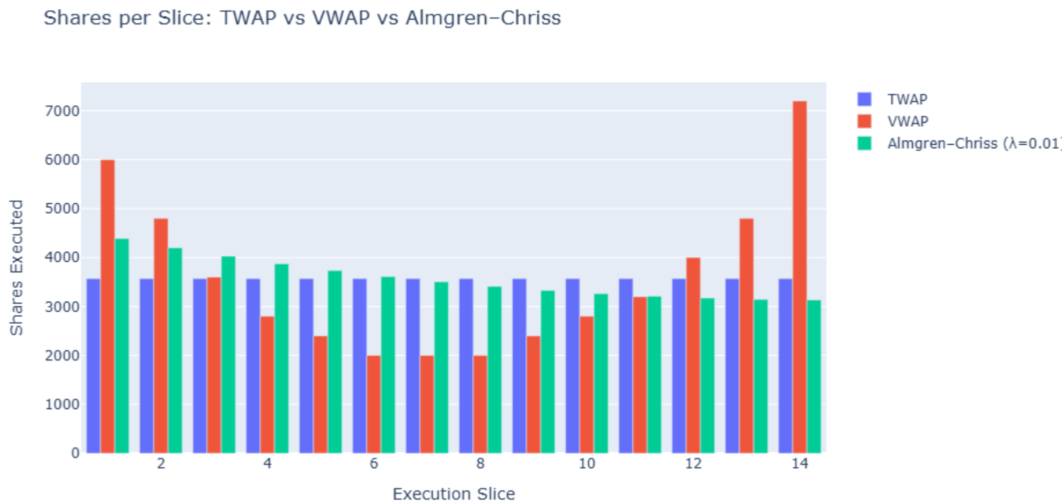

Figure 6: Comparison of shares executed per slice under TWAP, VWAP, and Almgren–Chriss.

The three strategies exhibit structurally different behaviors:

- **TWAP** follows a linear trajectory and allocates execution uniformly across time.
- **VWAP** aligns execution with expected liquidity, concentrating trades in high-volume periods.
- **Almgren–Chriss** balances cost and risk explicitly, generating a front-loaded execution path when $\lambda > 0$.

This deterministic comparison shows that execution is not only about *when* to trade, but also about *how aggressively* to reduce exposure.

6. Key Insights

1. TWAP is simple and transparent, but ignores both liquidity and risk.
2. VWAP is more market-aware, but remains heuristic.
3. Almgren–Chriss is the only strategy in this comparison that explicitly solves a cost–risk trade-off.
4. The urgency parameter λ plays a central role in shaping execution behavior.

7. Next Steps

The next stage extends the framework to a stochastic setting:

- Simulate price dynamics using Brownian motion:

$$S_{k+1} = S_k + \sigma\sqrt{\tau}\epsilon_k$$

- Incorporate market impact into execution prices:

$$P_k = S_k + \gamma \sum_{j<k} n_j + \eta \frac{n_k}{\tau}$$

- Evaluate performance through:
  - Implementation Shortfall
  - Expected Execution Cost
  - Execution Risk

  This will allow a full quantitative comparison of execution strategies under uncertainty.

8. References

- Almgren, R., & Chriss, N. (2001). Optimal execution of portfolio transactions.
- Standard execution benchmarks: TWAP and VWAP.
- Computational implementation and visualization developed in Python.